

Some Rigidity Criteria for Realization Spaces of Phased Matroids

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Consider the signs of maximal minors of a real full-rank matrix. These signs define a chirotope; hence a point of the real Grassmannian gives a chirotope up to relative sign pattern, which is called an oriented matroid. The realization space of an oriented matroid, equivalently the corresponding fiber in the real Grassmannian, can exhibit arbitrary semialgebraic complexity by Mnëv's universality theorem, a phenomenon often viewed as a form of Murphy's law in algebraic geometry.

In contrast, non-chirotopal phased matroids often display remarkable rigidity phenomena. Motivated by the work of Below, Krummeck, and Richter–Gebert, and by the subsequent rigidity results of Ruiz and Schaar, we study combinatorial conditions forcing realization rigidity for non-chirotopal phased matroids.

We prove that a realizable non-chirotopal phased matroid is projectively rigid whenever it contains a full-rank uniform restriction on $r + 2$ elements. Moreover, the bound $r + 2$ is sharp. This gives a sharp local combinatorial criterion for rigidity beyond the uniform setting of the results of Ruiz and Schaar.

Keywords: Grassmannian, Realization spaces, Phased matroids.